SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Poster Presentation

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO3 –Poster Presentation**	**Assessment:**	**Assessment Tool 1– Poster Presentation**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

STATISTICAL ANALYSIS OF EMPLOYEE SALARY DATA USING MEAN AND VARIANCE

A Statistical Approach to Understanding Salary Distribution and Employee Pay Patterns

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COURSE CODE: DSA0404
COURSE NAME:FUNDAMENTALS OF DATA SCIENCE

1 INTRODUCTION

Employee salary data provides valuable information about an organization's compensation structure. Mean and Variance summarize salary levels and show how widely salaries differ among employees.

Objective: Measure the average salary and degree of salary variation.

2 OBJECTIVES

- Calculate the Mean Salary
- Calculate the Variance
- Understand salary distribution
- Compare salaries with the average
- Support data-driven decisions

3 DATASET

Example Employee Salary Dataset

Employee	Salary (₹)
E1	25,000
E2	30,000
E3	35,000
E4	40,000
E5	45,000
E6	50,000
E7	55,000
E8	60,000

Employees: 8 **Variable: Monthly Salary**

4 METHODOLOGY

1. **Data Collection** — Collect salary records.
2. **Data Preparation** — Check missing or incorrect values.
3. **Calculate Mean** — Find the average salary.
4. **Calculate Variance** — Measure salary spread.
5. **Interpret Results** — Relate statistics to salary distribution.

MEAN FORMULA

$$\bar{x} = \Sigma x_i / n$$

x_i = salary of each employee | n = number of employees

POPULATION VARIANCE

$$\sigma^2 = \Sigma(x_i - \mu)^2 / N$$

μ = mean of data | N = total employees

5 RESULTS & SALARY DISTRIBUTION

MEAN SALARY	POPULATION VARIANCE	STANDARD DEVIATION
₹42,500	₹131,250,000	≈ ₹11,456

EMPLOYEE SALARY COMPARISON

6 INTERPRETATION

WHAT THE RESULTS MEAN

- ₹42,500 is the average monthly salary.
- Salaries range from ₹25,000 to ₹60,000.
- Variance indicates noticeable salary dispersion.
- Greater variance means greater salary differences.

Graph is kept only in the Results section.

7 KEY FINDINGS

Average	Range	Highest	Lowest	Variance
₹42,500	₹25k–₹60k	₹60,000	₹25,000	₹131.25M

Main finding: Mean shows the center; Variance shows the spread.

8 CONCLUSION

Mean and Variance provide a clear statistical summary of employee salary data. Mean identifies the typical salary level, while Variance measures salary variability. Together they support salary planning, compensation evaluation, budgeting, and workforce analysis.

MEAN → Average Salary | VARIANCE → Salary Spread

KEY TAKEAWAY

MEAN — Average Salary

VARIANCE — Salary Spread

STATISTICS — Better Decisions

A simple statistical approach for understanding employee pay patterns.